AI-Based Smart Expense Management System with Automated Categorization and Budget Recommendations

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***Abstract*—** Abstract— It is a challenging task to keep track of personal expenses as most of the expense management apps make extensive use of manual input. People tend to forget to enter their transactions and hence fail to provide complete record of their monthly expenses. This paper proposes an Android based intelligent expense management system that cuts down the manual effort required in keeping track of expenses by automatically extracting the expenses from the payment notifications that appear on the smartphone and also provides the option to manually enter the expenses when automatic extraction is not possible. In this system, a Multinomial Naive Bayes style classifier is used to classify the expenses and Linear Regression algorithm is used to predict the future monthly expenses from historical total expenses. The app is developed in Kotlin programming language and Jetpack Compose using SQLite for local database. The entire system is designed to run on-device so as to protect the sensitive financial data from getting transferred to any external server.

Keywords: Expense Management, Naive Bayes, Linear Regression, Android application, Personal Finance, Automated Classification, Expense Prediction

I. INTRODUCTION

Expense management is one of the aspects that help students, working professionals and individuals keep track of their monthly expenditure. In truth, however, there is a challenge when it comes to expense management as most of the current methods are dependent upon manual entries. People find that when they start using budgeting applications to help them keep track of their finances, the process of entering each of the transactions become difficult. In today's world, people rely on digital payments to make their transactions. Transactions through mobile wallets, UPI, debit card and other such modes result in a notification on the user's phone each time a transaction occurs. These notifications consist of information about the transaction amount and the name of the merchant. As such, it becomes possible for us to use this information for automatic recording of transactions. This helps us overcome the problem of manual efforts to a great extent as we can simply capture the information provided in the payment notification. There are some tools for expense management that provide a basic view of the spending done by the user in the past but these tools lack the ability to categorize intelligently.Similarly, without any prediction, the user would not be aware of their potential future spending. This project intends to solve this problem by designing an Android based expense management system that will be able to collect the transaction information from the payment notifications and also manual input of the expenses if they cannot be collected automatically. The system will make use of Naive Bayes algorithm to categorise the expenses and will use Linear Regression to predict the next month's spending based on the historical monthly totals. The contributions of this research are:

* An Android based application that can collect the transactions from the supported payment notifications and also manually enter the expenses;
* An automated categorisation of the transactions through Multinomial Naive Bayes style classifier; and
* The prediction of spending in the next month based on the historical monthly totals through Linear Regression.

The rest of this paper is organised as follows. In Section II, the related work is described. Section III deals with the methodology and the implementation of the system. Section IV deals with the current results, the evaluation and the limitations. Section V concludes the paper

II. RELATED WORK

A. Transaction Classification Text classification is one of the popular methods that can be utilized to label short text documents like messages, notifications, etc.

The Naive Bayes algorithm is very popular for text classification tasks because it is simple, efficient and applicable for small datasets. The effectiveness of probabilistic models for text classification task has been proved by McCallum and Nigam [1] who have compared different models for Naive Bayes text classification. The same issue has been studied by Manning et al. [2] within information retrieval and text classification frameworks.

When talking about expense management, transaction classification can be considered as a text classification task. The merchant's name and transaction description include tokens which help to detect what category transaction belongs to, whether it is food, travel, shopping, bills or some other. A classifier is much more flexible than the rule - based system which needs a set of keyword rules written manually. Instead of just keyword rules, a classifier is able to detect the pattern of each category from examples.

B. Spending Prediction

Forecasting is useful in personal finance because it helps users plan ahead rather than only reviewing past transactions. Regression models are commonly used for prediction problems where a relationship between an input variable and an output value needs to be estimated. Linear Regression is one of the simplest and most interpretable methods for this purpose. James et al. describe Linear Regression as a basic but effective supervised learning method for estimating continuous values [3].

Forecasting principles are also important when working with historical spending data. Hyndman and Athanasopoulos explain that predictions should be evaluated carefully, especially when future values are estimated from past observations [4]. In this project, monthly spending totals are used as historical data points, and Linear Regression is applied to estimate a future monthly expense value.

C. Research Gap

Existing expense tracking applications often focus on either recording transactions or displaying summaries. Some systems depend mostly on manual entry, while others use rigid rules for categorization. There is a need for a lightweight mobile application that combines automated transaction capture, machine learning-based categorization, local storage, and spending prediction in one workflow. The proposed system attempts to fill this gap by integrating these functions into an Android application.

III. METHODOLOGY

A. System Overview

The proposed system follows a modular architecture, as shown in Fig. 1. Transactions enter the system either through Android notification listening or manual entry. When a supported payment notification is received and its content is readable, the notification text is processed to extract the merchant name and transaction amount. If automatic extraction is not possible, the user can manually add the expense.

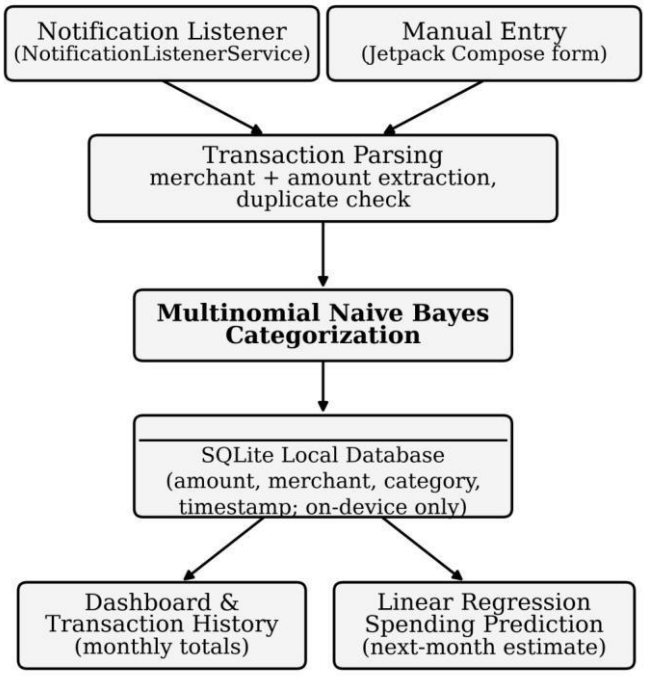


Fig. 1. Architecture of the proposed smart expense management system.

After parsing, the transaction is passed to the categorization module. The categorized transaction is then stored in the SQLite database along with the amount, merchant name, category, and timestamp. The dashboard displays the transaction history, monthly spending totals, and prediction results.

B. Method / Algorithm

The categorization module uses a Multinomial Naive Bayesstyle classifier. Each transaction is represented using text-based features extracted from merchant or transaction information. The classifier calculates the probability of each possible category and assigns the category with the highest probability.

$$P(c|x) \propto P(c) \times \prod P(x_i|c) \quad (1)$$

In (1), c represents a candidate expense category, x represents the features extracted from the transaction text, x_i represents an individual feature, $P(c)$ is the prior probability of category c, and $P(x_i|c)$ is the probability of observing feature x_i in category c.

For spending prediction, the system aggregates stored transactions into monthly totals. Linear Regression is then used to estimate a future monthly spending value based on previous monthly spending values.

$$y = \beta_0 + \beta_1 x + \varepsilon \quad (2)$$

In (2), y represents the predicted monthly spending, x represents the month index, β_0 is the intercept, β_1 is the slope coefficient, and ε represents the error term. The prediction is an estimate based on available historical data and should not be treated as a guaranteed value.

C. Implementation

The application is developed as a native Android mobile application using Kotlin. Jetpack Compose is used for designing the user interface. SQLite is used as the local database to store transaction records on the device. Android NotificationListenerService is used to read supported payment notifications when notification access permission is granted by the user.

The system includes duplicate handling to avoid saving the same transaction multiple times. This is done by checking whether a transaction with the same merchant and amount already exists within a recent time window.

TABLE I
SYSTEM COMPONENTS AND CONFIGURATION

Component	Description
Application platform	Android mobile application
Programming language	Kotlin
User interface	Jetpack Compose
Local database	SQLite
Notification integration	Android NotificationListenerService
Categorization algorithm	Multinomial Naive Bayes-style classifier
Prediction algorithm	Linear Regression
Duplicate handling	Merchant and amount matching within a recent time window

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The current version of the application has been implemented and functionally tested. The main tested functions include supported notification reading, manual expense entry, transaction parsing, duplicate checking, SQLite storage, category assignment, transaction history display, and prediction generation when sufficient monthly data is available.

A formal machine learning evaluation has not yet been completed. Therefore, this paper does not report accuracy, precision, recall, F1-score, confusion matrix, MAE, or RMSE values. For future evaluation, a labeled transaction dataset should be prepared. The Naive Bayes classifier can be tested using labeled merchant-category examples, while the Linear Regression model can be tested by comparing predicted monthly spending with actual monthly totals.

B. Results

The implemented system successfully provides the main functional outcomes expected from the project. It allows users to record expenses from supported payment notifications and through manual entry. It stores transactions locally with merchant name, amount, category, and timestamp. It also displays transaction history and generates spending predictions when enough monthly spending records are available.

Since no verified experimental dataset or measured metrics are currently available, the results are limited to implementation-based observations. The system should not be described as achieving a specific accuracy or prediction error until proper evaluation is completed.

C. Limitations

The system has some limitations. First, automatic transaction capture depends on whether the payment application exposes readable notification content. Therefore, the application cannot guarantee detection of every transaction from every payment application. Second, prediction results are only estimates based on historical monthly totals. They may be less reliable when the user has very limited transaction history. Third, the categorization model uses predefined merchant-category training examples, and its performance must be evaluated using a labeled dataset before making any accuracy claims.

V. CONCLUSION AND FUTURE WORK

In this paper, an intelligent expenditure management system using AI techniques for Android phones has been described. The proposed system makes the job of monitoring expenditures easy by fetching transaction information from payment notifications and through manual entry in case it cannot be done automatically. The proposed system makes use of the Naive Bayes classifier for classifying expenses and Linear Regression for predicting future expenditures. The app is developed using Kotlin, Jetpack Compose, and SQLite technologies. The proposed implementation successfully showcases the real-world applicability of transaction capture, machine learning for categorization, and spending prediction in a mobile application that helps users to manage their expenses. In the future, the work will concentrate on evaluating the proposed categorization and prediction algorithms. For Naive Bayes classifier, the following metrics should be considered – accuracy, precision, recall, F1-score, and confusion matrix. For the Linear Regression prediction algorithm, the following metrics should be used – MAE and RMSE, along with chronological testing. In future versions of the software, there might also be improvements related to budget recommendations, more diverse formats of notifications, and anomalies detection (unusual transactions)

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